Project Phase 2: ER Modeling

EER model of Team 54's WCA database

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Overview

We were assigned Team 54's (Apparatus) Phase 1 document. Their domain is the World Cube Association (WCA), which runs speedcubing competitions and keeps the official results, records and rankings. This document has our EER diagram for their mini-world (Section 1), our evaluation of their document (Section 2), the changes we made on top of that (Section 3), where we differ from the textbook notation (Section 4), our design quality analysis (Section 5) and how we used genAI (Section 6). Appendix A lists every entity and relationship as we drew it.

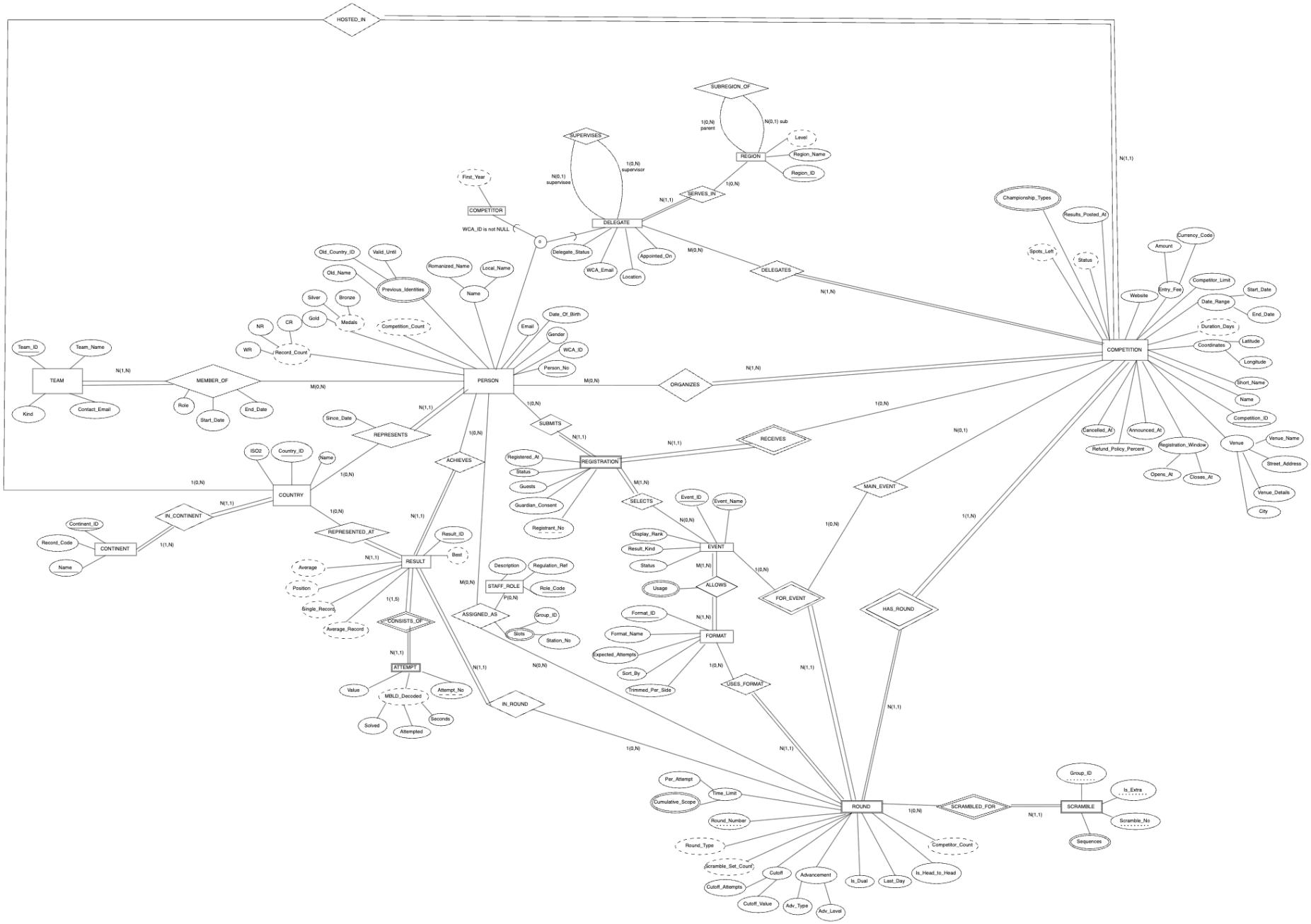
Page, table and constraint numbers like "Table 7, p. 8" or "C15" refer to Team 54's document.

1 EER diagram

The diagram is on the next page. It has all 16 entity types from Team 54's document (10 strong, 4 weak, and COMPETITOR and DELEGATE as subclasses of PERSON) and all 23 relationship types, with the fixes from Section 2 applied. It is exported as a vector PDF, so it can be zoomed. The diagram is also available here:

`https://drive.google.com/file/d/1WFcBBfIl0h5zeyIP1y9y-Ntq-mwySMaA/view?usp=sharing`

It uses the Elmasri & Navathe notation. Weak entities and identifying relationships have double borders, derived attributes are dashed ovals, multi-valued ones are double ovals, keys are underlined and partial keys have a dashed underline. Double lines mean total participation. Each edge has a (min,max) pair for the entity at that end, with the ratio letter next to it (see Section 4). The specialization of PERSON is overlapping ("o") and partial (single line), and COMPETITOR is predicate-defined by *WCA_ID IS NOT NULL*.


HOSTED_IN
SUBREGION_OF
1(0,N) parent
N(0,1) sub
REGION
Level
Region_Name
Region_ID
SUPERVISES
1(0,N) supervisor
N(0,1) supervisee
First_Year
SERVES_IN
1(0,N)
N(1,1)
COMPETITOR
DELEGATE
WCA_ID is not NULL
o
Delegate_Status
Appointed_On
WCA_Email
Location
M(0,N)
DELEGATES
N(1,N)
N(1,1)
Championship_Types
Results_Posted_At
Spots_Left
Status
Currency_Code
Amount
Entry_Fee
Competitor_Limit
Start_Date
Date_Range
End_Date
Website
Duration_Days
Coordinates
Latitude
Longitude
COMPETITION
Short_Name
Name
Competition_ID
Cancelled_At
Announced_At
Refund_Policy_Percent
Registration_Window
Opens_At
Closes_At
Venue
Venue_Name
Street_Address
Venue_Details
City
Old_Country_ID
Valid_Until
Old_Name
Previous_Identities
Romanized_Name
Local_Name
Name
Silver
Bronze
Gold
Medals
Competition_Count
NR
CR
WR
Record_Count
Date_Of_Birth
Email
Gender
WCA_ID
Person_No
PERSON
Team_ID
Team_Name
TEAM
Kind
Contact_Email
N(1,N)
MEMBER_OF
Role
Start_Date
End_Date
M(0,N)
ORGANIZES
N(1,N)
Since_Date
REPRESENTS
N(1,1)
1(0,N)
SUBMITS
N(1,1)
RECEIVES
1(0,N)
N(0,1)
ISO2
Country_ID
Name
COUNTRY
1(0,N)
ACHIEVES
Registered_At
Status
Guests
Guardian_Consent
Registrant_No
REGISTRATION
M(1,N)
SELECTS
Event_ID
Event_Name
N(0,N)
EVENT
Display_Rank
Result_Kind
Status
M(1,N)
ALLOWS
Usage
1(0,N)
FOR_EVENT
MAIN_EVENT
1(0,N)
HAS_ROUND
1(1,N)
Continent_ID
Record_Code
Name
CONTINENT
1(1,N)
IN_CONTINENT
N(1,1)
1(0,N)
REPRESENTED_AT
N(1,1)
Result_ID
Best
Average
Position
Single_Record
Average_Record
RESULT
1(1,5)
CONSISTS_OF
N(1,1)
Description
Regulation_Ref
STAFF_ROLE
Role_Code
P(0,N)
M(0,N)
ASSIGNED_AS
Group_ID
Slots
Station_No
Format_ID
Format_Name
Expected_Attempts
Sort_By
Trimmed_Per_Side
FORMAT
N(1,N)
1(0,N)
USES_FORMAT
N(1,1)
ATTEMPT
Value
MBLD_Decoded
Attempt_No
Solved
Attempted
Seconds
N(1,1)
N(0,N)
IN_ROUND
1(0,N)
N(1,1)
Per_Attempt
Cumulative_Scope
Time_Limit
Round_Number
Round_Type
Scramble_Set_Count
Cutoff_Attempts
Cutoff_Value
Cutoff
Adv_Type
Advancement
Adv_Level
Is_Dual
Last_Day
Is_Head_to_Head
Competitor_Count
ROUND
1(0,N)
SCRAMBLED_FOR
N(1,1)
Group_ID
Is_Extra
Scramble_No
SCRAMBLE
Sequences

2 Critical evaluation of Team 54's document

We read Sections 1–3 first to understand the domain, then went through every table in Section 4 and checked it against the rest of the document, mainly the functional requirements in Section 5 and the facts they quote from the WCA site. We sorted what we found into completeness (CP), correctness (CR) and modelability (MD) problems. Section 2.4 says which ones we fixed.

2.1 Completeness

CP1 Groups are used but never described. SCRAMBLE.Group_ID and the Slots on ASSIGNED_AS both refer to groups, and ROUND also stores Scramble_Set_Count, so the number of groups is stored twice and can disagree (Table 17, Table 15 on p. 11; relationship 19, p. 13).

CP2 C15 says SELECTS has a foreign key "into the competition's events", but there is no COMPETITION–EVENT relationship. The events a competition holds can only be worked out from ROUND. MAIN_EVENT has the same problem, since nothing says the headline event is actually held (C15, p. 15; relationships 6 and 11, p. 12).

CP3 Section 1.5 says a person's profile shows record counts (Max Park has 92 world records), but PERSON only has Competition_Count and Medals (p. 3 vs Table 7, p. 8).

CP4 "A trainee cannot delegate alone" (§1.1, U4) and "the supervisor is regional or senior, no cycles" (§4.6.3) are written in the text but are not in the constraint list C1–C20 (pp. 14–15).

CP5 TEAM.Contact_Email just has the domain "Address" (Table 11). Spots_Left is Competitor_Limit minus accepted registrations, which doesn't work when Competitor_Limit is NULL, and Table 10 allows that (p. 9).

CP6 D2 deletes "draft (never announced)" competitions and Table 4 has organizers creating drafts, but COMPETITION.Status only has cancelled, upcoming, in progress and past (p. 22, p. 5, p. 9).

CP7 There are spare scrambles for extra attempts (Is_Extra), but nothing records which scramble an attempt used (Tables 17 and 18).

2.2 Correctness

CR1 This was the main one. ACHIEVES connects RESULT to COMPETITOR, whose predicate is WCA_ID IS NOT NULL, with total participation on both sides. But Table 7 says a WCA_ID is only given once the person's first results are posted, and I2 enters attempts live during the competition. So during a newcomer's first competition their results have no valid owner (relationship 12, p. 13; Table 7, p. 8; I2, p. 20).

CR2 Section 1.6 says the year in a WCA ID is the year of the first competition, but C3 only checks that it is less than or equal to it (p. 4, p. 14).

CR3 Table 6 makes ISO2 a key "where one exists", so it can be NULL. Their own §4.6.1 then says ISO2 is declared UNIQUE NOT NULL. A candidate key can't be NULL anyway, and the WCA uses codes like XK where ISO doesn't have one (p. 7, p. 16).

CR4 RESULT.Average is "0 if the format has no average" (Table 14), but Table 1 says 333bf (Best of 5) also ranks Ao5 and 444bf/555bf (Best of 3) also rank Mo3.

CR5 I1 sets a registration to accepted or waiting_list automatically, but Table 4 says organizers handle "registration acceptance", and the pending status in Table 16 is never used (p. 20, p. 5, p. 11).

CR6 STAFF_ROLE includes delegate, so who delegated a competition can be stored in both DELEGATES and ASSIGNED_AS (Table 13, p. 10; relationships 17 and 19).

CR7 Round_Type is stored, but it follows from Round_Number and how many rounds the event has (the last one is the final), so the two can contradict each other (Table 15).

CR8 REGION.Level is top exactly when the region has no parent in SUBREGION_OF, so it repeats the relationship (Table 12; relationship 22).

CR9 ALLOWS.Usage holds one value, but some pairs like 333fm Best of 1 are both a full-round format and a cutoff format. C9 also never checks the cutoff format (relationship 8, p. 12; Table 1; C9, p. 14).

CR10 The document refers to Sections 2.1, 2.2, 2.3 and 3.3, which don't exist (Table 21, p. 17; §5.3, p. 20; C2, p. 14), and the paragraph in §1.3 is cut off at "refund policy (50" (p. 2).

CR11 Section 1.2 calls Head to Head a format, but it is stored as a flag on ROUND.

CR12 Entry_Fee and Refund_Policy_Percent are kept even though Table 3 puts payments and refunds out of scope.

2.3 Modelability

MD1 ROUND is a weak entity with two owners, COMPETITION and EVENT (key Competition_ID, Event_ID, Round_Number, like 333-r1). This is allowed but we had to look up how to draw it.

MD2 A cumulative time limit can cover several rounds, even of different events. That is really a relationship between ROUNDs, but they store it as the multi-valued attribute Cumulative_Scope (Table 15).

MD3 A staff assignment needs person, round, role, group and station together. We kept their ternary ASSIGNED_AS with the multi-valued composite Slots. Their §4.6.2 shows the other option, a weak entity ASSIGNMENT.

MD4 Previous_Identities.Old_Country_ID points at COUNTRY from inside an attribute, which can't be drawn as a link in ER (Table 7).

MD5 Whether a result is a record depends on every earlier result for that event, which ER has no way to show (Table 14).

MD6 MEMBER_OF is M:N, so there is only one instance per person and team. Someone who leaves and rejoins can't have two stints, even though there are Start_Date and End_Date attributes (relationship 23).

2.4 What we fixed

We fixed the things that make the model wrong or unclear, and kept the rest as Team 54 had it where a fix would add more than the mini-world needs. Most fixes are domain or constraint changes, so they don't show up in the diagram; the last column says where they do.

ID	Decision	What changed and why	In the diagram
CP1	Fix	Scramble_Set_Count is now derived from the distinct non-extra Group_IDs. We did not add a GROUP entity because groups are schedule logistics, which Team 54 put out of scope.	Dashed oval on ROUND

ID	Decision	What changed and why	In the diagram
CP2	Fix	Held events = events with a round 1 at that competition. C15 checks this with a trigger, and new C21 applies it to MAIN_EVENT.	—
CP3	Fix	Added Record_Count (derived, composite of WR, CR, NR) to PERSON.	Dashed composite on PERSON
CP4	Fix	New C22 (a trainee is never the only Delegate) and C23 (supervisor is regional or senior, no cycles).	—
CP5	Fix	Contact_Email must be a valid e-mail address. Spots_Left is NULL when there is no limit.	—
CP6	Fix	Status can also be draft (when Announced_At is NULL).	—
CP7	Keep	The WCA export doesn't link attempts to scrambles either.	—
CR1	Fix	ACHIEVES now connects PERSON (0,N) to RESULT (1,1). COMPETITOR stays as the subclass of people with a WCA ID.	ACHIEVES drawn from PERSON
CR2	Fix	C3 now says the WCA_ID year equals the first competition year.	—
CR3	Fix	ISO2 is NOT NULL.	ISO2 underlined
CR4	Fix	Average is computed for every 3- or 5-attempt round with all attempts present (except 333mbf), otherwise 0.	—
CR5	Fix	I1 creates a pending registration. The organizer accepts it, or it goes to the waiting list if Spots_Left = 0.	—
CR6	Fix	Removed delegate from STAFF_ROLE, so DELEGATES is the only place it is stored.	—
CR7	Fix	Round_Type is now derived.	Dashed oval on ROUND
CR8	Fix	Level is now derived from SUBREGION_OF.	Dashed oval on REGION
CR9	Fix	Usage is now multi-valued (a subset of full_round, cutoff_phase), and C9 also checks the cutoff format.	Double oval on ALLOWS
CR10	Fix	We read the references as Sections 4 and 5. We can't recover the rest of §1.3, so we used the attributes listed there and the 0–100 domain from Table 10.	—
CR11	Keep	Matches are out of scope, so a flag is enough.	—
CR12	Keep	These are facts published about a competition, not payment processing.	—

ID	Decision	What changed and why	In the diagram
MD1	Keep	Same key as WCIF round IDs.	Two double diamonds on ROUND
MD2	Keep	Rare case, not worth a 24th relationship. Checked by trigger (C11).	Double oval in Time_Limit
MD3	Keep	Ternary kept as Team 54 drew it.	Ternary with Slots
MD4	Keep	It becomes its own table with a foreign key to COUNTRY in Phase 3.	Plain attribute
MD5	Keep	Record markers are stored and recomputed by a trigger.	Dashed ovals on RESULT
MD6	Keep	Only the latest stint is stored. None of Team 54's reports need history.	—

3 Changes beyond the evaluation

None. Everything we changed is one of the fixes in Section 2.4. We didn't add, remove or rename any entity or relationship.

To check this, we went through Tables 5–18 and the relationship list in §4.3 one row at a time. Team 54 wrote participation as T or P and ratios as 1:N or M:N. We turned T into a minimum of 1 and P into a minimum of 0, took the maximum from the ratio, and used tighter limits where they gave one (1 to 5 attempts per result, at most one supervisor). Format_Name and WCA_Email are described as unique but Team 54 lists them as simple attributes, so we didn't underline them.

There are two things we left as they are on purpose:

1. Because of CP6 a competition can now be a draft. A draft has no rounds, Delegate or organizer yet, but HAS_ROUND, DELEGATES and ORGANIZES are (1,N) on the COMPETITION side. Team 54's C6 already says this is "checked when announced", so we kept (1,N) and will check it when the competition is announced instead of on insert.
2. Some rules can't be shown with (min,max): at most 4 rounds per event (C7), one registration per person per competition, C22 and C23. These stay as constraints.

4 Notational deviations

#	Deviation	Reason
1	Ratio letters next to (min,max)	We kept the 1/N/M labels so the diagram is easier to read. They read the opposite way from (min,max), which is why COUNTRY has "N (1,1)" on IN_CONTINENT. The (min,max) pair is the actual constraint.
2	Some alternate keys not drawn	Single-attribute alternate keys are underlined (ISO2, Event_Name, CONTINENT.Name). The ones made of other entities' keys, RESULT's (Competition_ID, Event_ID, Round_Number, Person_No) and REGISTRATION's (Competition_ID, Person_No), can't be drawn, and WCA_ID is only unique when not NULL. They are in Appendix A.
3	Old_Country_ID is a plain attribute	ER can't link an attribute to an entity. The foreign key to COUNTRY is added in Phase 3.

#	Deviation	Reason
4	Slots is a multi-valued attribute on a relationship	The textbook allows attributes on relationships but has no multi-valued example, so we drew a double oval on the ASSIGNED_AS diamond.
5	Some constraints not drawn	Rules like at most 4 rounds per event can't be written with (min,max). See Section 3.
6	Colours	The fills and red labels are only for readability and don't mean anything.

5 Design quality analysis

5.1 Redundancy

RESULT stores Best, Average, Position and the two record markers, which can all be worked out from its ATTEMPT rows. Team 54 keep them stored on purpose because rankings are computed over millions of results (§4.5). We drew them as derived, but in the database they will be stored twice.

The year someone first competed is in three places. After our CR2 fix it has to be the first four digits of WCA_ID. COMPETITOR.First_Year is derived from the same digits. And it can also be found by going from PERSON through ACHIEVES, RESULT, ROUND and COMPETITION to the earliest Start_Date. If the Results Team merges two person records and the older results came from an earlier year, the year in the WCA_ID no longer matches.

DELEGATE.Location holds a string like "USA (California)" and the region the Delegate is linked to through SERVES_IN has Region_Name "USA (California)". That is the same thing stored twice. We also have country history twice: Previous_Identities has Old_Country_ID and Valid_Until, and REPRESENTED_AT records the country on every result. Finally, Group_ID appears both as part of SCRAMBLE's key and inside the Slots on ASSIGNED_AS, and nothing stops someone being assigned to group D of a round that has no group D.

REPRESENTS and REPRESENTED_AT look like a duplicate, but they aren't, since one is the country now and the other is the country on the day of the result.

5.2 Update anomaly

Yes. Take Team 54's U1, where a score taker types 812 instead of 1812 for someone's third attempt in 333-r1. Fixing that one ATTEMPT.Value means changing that RESULT's Best and Average, the Position of every other result in 333-r1, and, if 8.12 had been marked as a national record, the record markers on this result and on later ones. If the person no longer qualifies, their results in 333-r2 also have to be deleted. Missing any of these leaves the rankings out of line with the attempts, so in Phase 3 all of it will be done by one trigger on ATTEMPT and nobody can edit those columns directly.

Renaming a region is a smaller example. If "USA (California)" is renamed, every DELEGATE.Location with the old name has to be changed too. We plan to drop Location and get the name through SERVES_IN.

5.3 Insertion anomaly

Team 54 say retired events like 333ft and magic still have to be in EVENT for old records (§1.2). But EVENT has total participation in ALLOWS, and Table 1 only lists formats for current events. So to add a retired event and its old results we would first have to make up which formats it is allowed to use. In Phase 3 we will only require ALLOWS for official events.

Two smaller cases come from the same kind of (1,N) rule. A draft competition can't be saved until it has a Delegate, an organizer and a round, even though organizers start drafts before a Delegate is found (Table 4). And a new TEAM can't be added until it has at least one member.

5.4 Deletion anomaly

Team 54's D1 deletes a withdrawn REGISTRATION and also deletes that person's ASSIGNED_AS rows in the competition. But a person can withdraw from competing and still judge. Their judging assignments are lost, and the J4 staffing report counts them wrong. We will remove that cascade.

There is no "former" value for Delegate_Status, so the only way to show that a Delegate has retired is to delete their DELEGATE row. If that delete cascaded, the DELEGATES rows for every competition they ran would go with it, and we would lose who delegated past competitions like WC2025. Under Team 54's default ON DELETE RESTRICT (C2) the delete is refused instead, so the person has to stay a Delegate with a status that is no longer true. We will add a former status instead of deleting.

MEMBER_OF has the same issue from MD6. If someone leaves a team and the row is deleted, there is no record they were ever a member.

6 Use of generative AI

We used Claude to check our evaluation against Team 54's document and find the page numbers for each point, to compare the diagram labels with our inventory table, to help write Section 5 from our notes, and to format the LaTeX. The modelling decisions are our own.

A Inventory of entities and relationships as modelled

All 16 entity types and 23 relationship types as drawn, with the fixes applied. K = key, PK = partial key, C = composite, MV = multi-valued, D = derived, S = simple, AK = alternate key. For relationships, (min,max) gives, for each participant, how many relationship instances one entity takes part in (Elmasri §3.7.4).

#	Name	Kind	Attributes (kind)	Keys	Fix applied
E1	CONTINENT	Strong	Continent_ID (K); Name (K); Record_Code (S)	PK Continent_ID; AK Name	Kept
E2	COUNTRY	Strong	Country_ID (K); ISO2 (K); Name (S)	PK Country_ID; AK ISO2	ISO2 NOT NULL (CR3)
E3	PERSON	Strong (superclass)	Person_No (K); WCA_ID (S, unique); Name (C: Romanized_Name, Local_Name); Gender (S); Date_of_Birth (S); Email (S); Previous_Identities (MV, C: Old_Name, Old_Country_ID, Valid_Until); Competition_Count (D); Medals (D, C: Gold, Silver, Bronze); Record_Count (D, C: WR, CR, NR)	PK Person_No; WCA_ID unique when not NULL	Record_Count added (CP3); WCA_ID year rule (CR2)
E4	COMPETITOR	Subclass of PERSON (overlapping, partial)	First_Year (D); inherits all of PERSON. Predicate: WCA_ID IS NOT NULL	PK Person_No (inherited)	No longer the ACHIEVES participant (CR1)
E5	DELEGATE	Subclass of PERSON (overlapping, partial)	Delegate_Status (S); WCA_Email (S, unique); Location (S); Appointed_On (S); inherits all of PERSON	PK Person_No (inherited); WCA_Email unique	Trainee and supervision rules C22, C23 (CP4)
E6	EVENT	Strong	Event_ID (K); Event_Name (K); Display_Rank (S); Result_Kind (S); Status (S)	PK Event_ID; AK Event_Name	Kept
E7	FORMAT	Strong	Format_ID (K); Format_Name (S, unique); Expected_Attempts (S); Sort_By (S); Trimmed_Per_Side (S)	PK Format_ID; Format_Name unique	Kept

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#	Name	Kind	Attributes (kind)	Keys	Fix applied
E8	COMPETITION	Strong	Competition_ID (K); Name, Short_Name (S); Venue (C: Venue_Name, Street_Address, Venue_Details, City); Coordinates (C: Latitude, Longitude); Date_Range (C: Start_Date, End_Date); Registration_Window (C: Opens_At, Closes_At); Entry_Fee (C: Amount, Currency_Code); Competitor_Limit (S); Refund_Policy_Percent (S); Website (S); Announced_At, Cancelled_At, Results_Posted_At (S); Championship_Types (MV); Duration_Days (D); Status (D); Spots_Left (D)	PK Competition_ID	*draft* status (CP6); Spots_Left NULL if no limit (CP5)
E9	TEAM	Strong	Team_ID (K); Team_Name (S); Kind (S); Contact_Email (S)	PK Team_ID	E-mail domain (CP5)
E10	REGION	Strong	Region_ID (K); Region_Name (S); Level (D)	PK Region_ID	Level derived (CR8)
E11	STAFF_ROLE	Strong	Role_Code (K); Description (S); Regulation_Ref (S)	PK Role_Code	*delegate* removed (CR6)
E12	RESULT	Strong	Result_ID (K); Best (D); Average (D); Position (D); Single_Record, Average_Record (D)	PK Result_ID; AK (Competition_ID, Event_ID, Round_Number, Person_No)	Average rule (CR4)
E13	ROUND	Weak (owners COMPETITION, EVENT)	Round_Number (PK); Round_Type (D); Time_Limit (C: Per_Attempt, Cumulative_Scope (MV)); Cutoff (C: Cutoff_Attempts, Cutoff_Value); Advancement (C: Adv_Type, Adv_Level); Scramble_Set_Count (D); Is_Dual, Is_Head_to_Head (S); Last_Day (S); Competitor_Count (D)	(Competition_ID, Event_ID, Round_Number)	Round_Type derived (CR7); Scramble_Set_Count derived (CP1)
E14	REGISTRATION	Weak (owner COMPETITION)	Registrant_No (PK); Status (S); Registered_At (S); Guests (S); Guardian_Consent (S)	(Competition_ID, Registrant_No); AK (Competition_ID, Person_No)	*pending* used (CR5)

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#	Name	Kind	Attributes (kind)	Keys	Fix applied
E15	SCRAMBLE	Weak (owner ROUND)	Group_ID (PK part); Is_Extra (PK part); Scramble_No (PK part); Sequences (MV)	(Competition_ID, Event_ID, Round_Number, Group_ID, Is_Extra, Scramble_No)	Group_ID stands for groups (CP1)
E16	ATTEMPT	Weak (owner RESULT)	Attempt_No (PK); Value (S); MBLD_Decoded (D, C: Solved, Attempted, Seconds)	(Result_ID, Attempt_No)	Kept

#	Name	Meaning	Kind	Attributes	Ratio	(min,max) per side	Fix applied
R1	IN_CONTINENT	each country lies in one continent	Binary	—	N:1	COUNTRY (1,1); CONTINENT (1,N)	Kept
R2	REPRESENTS	current citizenship country (Reg. 2e)	Binary	Since_Date (S)	N:1	PERSON (1,1); COUNTRY (0,N)	Kept; change rule C16
R3	HOSTED_IN	one host country per competition	Binary	—	N:1	COMPETITION (1,1); COUNTRY (0,N)	Kept
R4	HAS_ROUND	a competition holds $\geq$1 round	Identifying	—	1:N	COMPETITION (1,N); ROUND (1,1)	Kept; checked at announcement (C6)
R5	FOR_EVENT	each round is of one event; $\leq$4 per competition–event	Identifying	—	1:N	EVENT (0,N); ROUND (1,1)	Kept (MD1)
R6	MAIN_EVENT	optional headline event	Binary	—	N:1	COMPETITION (0,1); EVENT (0,N)	Must be a held event, C21 (CP2)
R7	USES_FORMAT	each round has one format	Binary	—	N:1	ROUND (1,1); FORMAT (0,N)	Format allowed, C9
R8	ALLOWS	format table of Reg. 9b	Binary	Usage (MV) $\subseteq$ {full_round, cutoff_phase}	M:N	EVENT (1,N); FORMAT (1,N)	Usage made MV (CR9)
R9	RECEIVES	registrations exist only in a competition	Identifying	—	1:N	COMPETITION (0,N); REGISTRATION (1,1)	Kept
R10	SUBMITS	$\leq$1 registration per person per competition	Binary	—	1:N	PERSON (0,N); REGISTRATION (1,1)	Kept; enforced by AK of E14

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#	Name	Meaning	Kind	Attributes	Ratio	(min,max) per side	Fix applied
R11	SELECTS	chosen events; only held ones	Binary	—	M:N	REGISTRATION (1,N); EVENT (0,N)	Held events via ROUND, C15 (CP2)
R12	ACHIEVES	every result belongs to one person	Binary	—	1:N	PERSON (0,N); RESULT (1,1)	COMPETITOR → PERSON (CR1)
R13	IN_ROUND	≤1 result per round and person	Binary	—	1:N	ROUND (0,N); RESULT (1,1)	Kept
R14	REPRESEN-TED_AT	country on the day of the result	Binary	—	N:1	RESULT (1,1); COUNTRY (0,N)	Kept
R15	CONSISTS_OF	1–5 attempts, never above the format	Identifying	—	1:N	RESULT (1,5); ATTEMPT (1,1)	Kept; C10
R16	SCRAMBLED_FOR	scrambles per group of a round	Identifying	—	1:N	ROUND (0,N); SCRAMBLE (1,1)	Groups via Group_ID (CP1)
R17	DELEGATES	≥1 WCA Delegate per competition (Reg. 1a)	Binary	—	M:N	DELEGATE (0,N); COMPETITION (1,N)	Sole Delegate source (CR6); C22
R18	ORGANIZES	≥1 organizer per competition (Reg. 1a)	Binary	—	M:N	PERSON (0,N); COMPETITION (1,N)	Kept
R19	ASSIGNED_AS	e.g. judge in group B, station 6	Ternary	Slots (MV, C: Group_ID, Station_No)	M:N:P	PERSON (0,N); ROUND (0,N); STAFF_ROLE (0,N)	No delegate role (CR6); ternary kept (MD3)
R20	SUPERVISES	non-senior Delegates have one supervisor	Recursive	—	1:N	DELEGATE as supervisor (0,N); DELEGATE as supervisee (0,1)	C23 (CP4)
R21	SERVES_IN	a Delegate belongs to one region	Binary	—	N:1	DELEGATE (1,1); REGION (0,N)	Kept
R22	SUBREGION_OF	e.g. “USA (California)” in a top region	Recursive	—	N:1	REGION as sub (0,1); REGION as parent (0,N)	Now derives Level (CR8)
R23	MEMBER_OF	team, committee and Board membership	Binary	Role (S); Start_Date (S); End_Date (S)	M:N	PERSON (0,N); TEAM (1,N)	Kept; latest stint only (MD6)